

Cboe Titanium Cboe Canada Equities FIX Specification

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Introduction

Overview

This specification is the standard FIX specification to be used for the Cboe Canada Inc. (Cboe Canada) ECN platform within the TR2 data center. Cboe Canada comprises two trading platforms, which are as follows:

- the NEO-L, NEO-N, and NEO-D Trading Books, plus the Special Settlement Term order book and the Crossing Facility (collectively, Cboe Canada NEO); and
- the MATCHNow Trading Book (MATCHNow).

Cboe Canada Members use a subset of the FIX 4.2 protocol for order entry and drop copies.

It is assumed the reader is familiar with the FIX 4.2 protocol as described at http://www.fixprotocol.org. This document describes the differences between the Cboe Canada implementation and the FIX 4.2 standard.

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Certification Requirement

All customers must complete a formal certification in the appropriate Cboe Certification test environment before production orders will be accepted by Cboe Canada. Formal certification scripts can be found in the Cboe Customer Web Portal. Customers may complete the formal certification using the Certification Tool app and selecting the applicable certification script. Customers are advised to test all functionality they plan to use in production in the Cboe Certification test environment.

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Data Types

Timestamps

All FIX timestamps are GMT as per the FIX standard. Members are expected to synchronize their clocks with an external time source. Timestamps from Cboe Canada may be in microsecond YYYYMMDD-HH:MM:SS.ssssss or nanosecond YYYYMMDD-HH:MM:SS.sssssssss precision, controlled by the per-port settings. Timestamps sent to Cboe Canada may be in any precision but will be processed to no more than microsecond precision.

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Prices

Members should program their systems to allow execution prices to be returned with up to four decimals. The currency (CAD or USD) will be determined by the symbol being traded.

Table 1. MATCHNow Prices

MIN VALUE	MAX VALUE	TICK SIZE
0.0001	0.50	0.0001
0.50	Max allowable price	0.001

Table 2. Cboe Canada NEO Prices - Midpoint Orders

MIN VALUE	MAX VALUE	TICK SIZE
0.005	0.50	0.0025
0.50	Max allowable price	0.005

Table 3. Cboe Canada NEO Prices - All Other Orders

MIN VALUE	MAX VALUE	TICK SIZE
0.005	0.50	0.005
0.50	Max allowable price	0.01

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Hours of Operation

Refer to the website for the Cboe Canada holiday schedule and market hours.

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Protocol Features

The trading platform does not guarantee messages sent by Members to the trading platform, including through protocols such as TCP. Members are responsible to monitor the status of the messages they send to the trading platform.

Maximum Open Order Limits

The Exchange limits the maximum number of open orders allowed on a FIX port to 100,000 per port. New orders will be rejected once this limit is breached until the number of open orders drops back below 100,000.

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Protected NBBO

The Canadian market distinguishes between the National Best Bid and Offer (NBBO) and the Protected National Best Bid an Offer (PNBBO). PNBBO refers to the best bid and offer of protected prices across all national venues.

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Identifiers

ClOrdID

Day-unique ID chosen by client. 20 characters or less for single new orders, 18 characters or less for cross order entry. All characters in ASCII range 32-126 are allowed, excluding tilde (~), comma (,), semicolon (;), and pipe (|). Note the excluded character ~ may be used by Cboe Canada in cases where a *ClOrdID* is produced by Cboe Canada.

If the *ClOrdId* matches a live order it will be rejected as duplicate (unless *PossResend* (97) =Y).

Note: Cboe Canada only enforces the uniqueness of *ClOrdID* values among currently live orders. However, it is strongly recommended that firms use unique *ClOrdID* values for the day.

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OrderID

Day-unique id assigned by Cboe Canada to an order (and to each side of a cross). They are 12-character base-36 upper-case alpha-numeric strings.

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ExecID

Identifier of an execution message. Day-unique within MATCHNow and all Cboe Canada NEO (an ExecID assigned by MATCHNow may be duplicated in the Cboe Canada NEO).

Note: ***ExecIDs*** are represented on ODROP and FIXDROP ports as nine-character, upper-case base 36 ASCII. Leading zeros should be added if the converted base 36 value is shorter than nine characters.

Table 4. ExecID Example Conversion

DECIMAL	BASE 36
28294005440239	A1234B567
76335905726621	R248BC23H
728557228187	09AP05V2Z

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FIX Session Logical Connectivity and Routing

MATCHNow

MATCHNow operates a separate logical environment. Therefore, order entry FIX sessions and drop sessions are provisioned separately from Cboe Canada NEO.

- A FIX port (session) must be provisioned specifically for MATCHNow to interact with the MATCHNow book.
- Drop ports must be provisioned specifically for MATCHNow to receive drop traffic for MATCHNow.

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Cboe Canada NEO

- A Cboe Canada FIX port (session) can access any Cboe Canada trading book **except** for MATCHNow.
- Order flow can be directed to all non-MATCHNow Cboe Canada trading books via the *ExDestination* (100) field.
- Drop ports provisioned for Cboe Canada NEO will not receive MATCHNow traffic.

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Order Restatements

There are several scenarios where Cboe Canada will send a restatement of an order to inform the client of an important update to the order. Notable FIX tag values on these restatements are as follows:

- Good 'Til "Done For Day"
  - Issued at the end of the trading day for any unterminated Good 'Til Cancel (GTC) and Good 'Til Date (GTD) orders.
  - *ExecTransType* = 3 (Status), *ExecType* = 3 (DoneForDay), *ExecRestatementReason* = 1 (Good'Til restatement)
  - *OrdStatus* = 3 (DoneForDay)
- Good 'Til Carried Order Restatement
  - GTC and GTD orders will be restated at the start of each business day, after FIX connection establishment and before regular trading activity messages.
  - *ExecTransType* = 3 (Status), *ExecType* = D (Restated), *ExecRestatementReason* = 1 (Good'Til restatement)
- Order Protection (OPR) and Market-to-Limit (MTL) pricing
  - *ExecTransType* = 3 (Status), *ExecType* = D (Restated), *ExecRestatementReason* = 3 (Repricing)
  - *Price* provides the updated price of the order
- Iceberg display size refreshed
  - *ExecTransType* = 3 (Status), *ExecType* = D (Restated), *ExecRestatementReason* = 4 (Iceberg refreshed)
  - *SecondaryOrderID* provides the order id used in market data to display the reloaded iceberg.
  - *DisplayQty* provides the new size displayed for the order
- Partial reduce due to self-trade prevention
  - *ExecTransType* = 3 (Status), *ExecType* = D (Restated), *ExecRestatementReason*= 5 (Reduce)
  - *OrderQty* provides the new order quantity
- Triggered stop order
  - *ExecTransType* = 3 (Status), *ExecType* = D (Restated), *ExecRestatementReason* = 18 (Triggered Stop)

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Protocol

Message Format

FIX messages are ASCII formatted. Members will be provided with a *SenderCompId* and *SenderSubId* that must be sent on every message. The *TargetCompId* for all messages the Member sends must be MNOW or NEO.

All messages the Member receives will have the Sender and Target fields swapped.

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Sequence Numbers

Sequence numbers, both inbound and outbound, will be reset to 1 each night during the down time.

Messages are processed in sequence order. Behind sequence messages (other than **`Sequence Reset`** message- Reset) cause immediate logout. Ahead of sequence messages (other than a **`Resend Request`** message) trigger a message recovery via a **`Resend Request`** message.

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Version Compatibility

Cboe uses the FIX 4.2 session protocol.

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Sessions

The following session messages are supported in both directions:

Table 5. Session Messages Supported in Both Directions

MESSAGE	TYPE	COMMENT
Logon	A	Begin session (or resume a broken session).
Heartbeat	0	
Test Request	1	
Resend Request	2	
Reject	3	Malformed message or improper session level handling.
Sequence Reset	4	Both Gap Fill (GapFillFlag = Y) and Reset.
Logout	5	Used to gracefully close session.

Connectivity and Certification

Table 6. Connectivity and Certification

ATTRIBUTE	DESCRIPTION	INPUT
IP Address	Address to connect to.	Supplied by Cboe.
TCP Port	Port to connect to.	Supplied by Cboe.
SenderCompID	Sent in every FIX message to Cboe.	Supplied by Cboe.
SenderSubID	Sent in every FIX message to Cboe.	Supplied by Cboe.
TargetCompID	Sent in every FIX message to Cboe.	MNOW for MATCHNow connections NEO for all Cboe Canada NEO connections.
TargetSubID	Sent in every FIX message to Cboe.	TEST for test/certification system. PROD for production.

For information on connectivity options to Cboe Canada, refer to the Cboe Canada Connectivity Manual.

All new Members must pass certification before they will be permitted to enter orders in the production marketplace. Certain new order types and options will also require certification before they may be utilized. Contact the Cboe Canada Trade Desk for more information.

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Logon Message Fields

Table 7. Logon Message Fields

TAG	FIELD NAME	REQUIRED	DESCRIPTION
35	*MsgType*	Y	A
108	*HeartbeatInterval*	Y	Client Heartbeat Interval (in seconds).

`Logon` message must be the first message sent by the Member after the TCP connection is established. EncryptMethod is ignored (FIX level encryption is not supported).

Cboe Canada will wait one second after a Logon is received to ensure that no **`Resend Request`** messages are in flight from the Member. A **`Heartbeat`** message will be sent to indicate that the one second wait period has ended. **Members should not send any orders prior to receiving this first Heartbeat from Cboe Canada.**

The IP Address of the Member, the *SenderCompId*, *SenderSubId* and *TargetCompId* (defined in the Logon Message Fields Table) and *TargetSubId* (TEST/PROD) will be validated. If validation fails, the connection will be dropped without a reject (to avoid corrupting the Member's sequence in the case the Member mistakenly connected to the wrong port).

If connection is unexpectedly broken, upon reconnection the Member may receive a login reply with a sequence number greater than expected. This means in-flight messages were missed (likely important Execution Reports). The Member should issue a **`Resend Request`** message to retrieve the missed messages.

Similarly, Cboe Canada will issue a **`Resend Request`** message to the Member for messages that it missed. The Member may wish to send gap fill messages in place of new orders to avoid re-submission of potentially stale orders.

Members must specify *HeartbeatInterval* in the **`Logon`** message. This value will be clamped between 5 and 300 seconds and be returned in the logon reply message. Cboe Canada recommends using as low a value as the reliability and latency of your connectivity will allow.

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Heartbeat Message Fields

Table 8. Heartbeat Message Fields

TAG	FIELD NAME	REQUIRED	DESCRIPTION
35	*MsgType*	Y	0
112	*TestReqID*	N	Required in response to a Test Request.

A **`Heartbeat`** message should be sent if the agreed upon *HeartbeatInterval* has elapsed since the last message sent. If any message was sent during the preceding *HeartbeatInterval,* a **`Heartbeat`** message does not need to be sent.

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Test Request Message Fields

Table 9. Test Request Message Fields

TAG	FIELD NAME	REQUIRED	DESCRIPTION
35	*MsgType*	Y	1
112	*TestReqID*	Y	Auto-generated request ID.

If a ***HearbeatInterval*** + 1 seconds have elapsed since the last message received, a **`Test Request`** message should be issued. If another ***HearbeatInterval*** + 1 seconds elapse without receiving a message, the TCP connection should be dropped. This ensures a broken TCP connection is detected even if the TCP stack doesn't notice (this has been observed to happen in WAN environments, particularly when a VPN is involved).

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Resend Request Message Fields

Table 10. Resend Request Message Fields

TAG	FIELD NAME	REQUIRED	DESCRIPTION
35	*MsgType*	Y	2
7	*BeginSeqNo*	Y	Sequence number of first message in range to be resent.
16	*EndSeqNo*	Y	0 means + infinity

A **`Resend Request`** message should be processed even if it is received ahead of sequence. Only after resending the requested range (all marked *PossDup*= Y, including any gap fills) should **`Resend Request`** message be issued in the opposite direction.

As discussed in the FIX 4.2 specification, it is possible to send an open or closed sequence range in a **`Resend Request`** message (an open range uses sequence zero as the *EndSeqNo*). Cboe Canada will honor either type of request but will always issue **`Resend Request`** messages with a closed sequence range.

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Reject Message Fields

Table 11. Reject Message Fields

TAG	FIELD NAME	REQUIRED	DESCRIPTION
35	*MsgType*	Y	3
45	*RefSeqNum*	Y	*MsgSeqNum* of rejected message.
371	*RefTagID*	N	
372	*RefMsgType*	N	
373	*SessionRejectReason*	N	
58	*Text*		

Session level rejects indicate violations of the session protocol or missing (or bogus) fields. These are to be expected during development and certification while the Member is being adapted for Cboe Canada but should be extremely rare in production. Application layer rejects such as order rejection (35=D, 39=8) or cancel rejects (35=9) are normal.

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Sequence Reset Message Fields

Table 12. Sequence Reset Message Fields

TAG	FIELD NAME	REQUIRED	DESCRIPTION
35	*MsgType*	Y	4
36	*NewSeqNo*	Y	Next expected sequence number.
123	*GapFillFlag*	N	Sequence Reset - Gap Fill messages (*GapFillFlag*= Y) must be received in sequence. Any messages (including any Gap Fills) sent in response to a Resend Request should have *PossDup*= Y. Sequence Reset - Reset (*GapFillFlag* not Y) is used only as a last resort, and always by human intervention, to allow an otherwise hopelessly confused session to be resumed. In these cases, all chances at automatic message recovery are lost.

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Logout Message Fields

Table 13. Logout Message Fields

TAG	FIELD NAME	REQUIRED	DESCRIPTION
35	*MsgType*	Y	5
58	*Text*	N	Indicates reason for Logout.

Either side may issue a **`Logout`** message to gracefully close the session. The side issuing the **`Logout`** message should process messages normally until it sees the logout reply, and then break the TCP connection. Cboe Canada typically only requests logout after the scheduled end of FIX session.

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FIX Messages

Standard Message Header Message Fields

Table 14. Standard Message Header Message Fields

TAG	FIELD NAME	REQUIRED	DESCRIPTION
8	*BeginString*	Y	FIX.4.2 Must be first field in message.
9	*BodyLength*	Y	Length of message following *BodyLength* field up to and including the delimiter preceding the *CheckSum* field. Must be second field in message.
35	*MsgType*	Y	Must be third field in message. 0 = Heartbeat 1 = TestRequest 2 = Resend/Request 3 = Reject 4 = SequenceReset 5 = Logout 8 = ExecutionReport 9 = OrderCancelReject A = Logon D = NewOrderSingle F = OrderCancelRequest G = OrderCancelReplaceRequest UCC= Trade Cancel/Correct
49	*SenderCompID*	Y	ID of sender: Assigned by Cboe for messages sent to Cboe Canada (4 character upper alphanumeric). (*TargetCompID* for messages from Cboe Canada)
50	*SenderSubID*	Y	Sub ID of sender: Assigned by Cboe Canada for messages sent to Cboe Canada. (4 digit numeric) (*TargetSubID* for messages from Cboe Canada)
56	*TargetCompID*	Y	ID of destination (MNOW or NEO for messages sent to MATCHNow or Cboe Canada NEO markets, respectively) (*SenderCompID* for messages from Cboe Canada)
57	*TargetSubID*	Y	Sub ID of destination: TEST for messages sent to Cboe Canada test/certification system. PROD for messages sent to Cboe Canada production system. (*SenderSubID* for messages from Cboe Canada)
34	*MsgSeqNum*	Y	Sequential sequence number for session.
43	*PossDupFlag*	N	N = (Default) Y = Indicates a message resend and should be used during FIX Replay. All inbound orders with a value of 'Y' will be rejected by Cboe.
52	*SendingTime*	Y	GMT date-time that message was sent.
122	*OrigSendingTime*	N	For messages with *PossDupFlag* =Y, indicates time that message was first sent.
115	*OnBehalfOfCompId*	Y	Identifies the broker. Provided on every message.

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TAG	FIELD NAME	REQUIRED	DESCRIPTION
			Must be the Broker ID of the broker responsible for the order (3-digit numeric identifier, with leading zeros when necessary).
128	*DeliverToCompId*	N	Identifies the broker (value sent in *OnBehalfOfCompId*). Used by Cboe on return messages back to client according to standard FIX 4.2 Errata.
97	*PossResend*	N	**New Order Single (35=D)** N = (Default) indicates a new order. Y = Indicates an application level resend and is not supported. The message will be logged, but effectively ignored. For reasons of economy, Cboe Canada does not track (in primary storage) the *ClOrdID* values of orders that are no longer live. For reasons of performance, Cboe Canada does not access secondary storage to enforce unique *ClOrdID* values against orders that are no longer live. Without full duplicate *ClOrdID* value enforcement, it is not possible to safely implement the full behavior specified in the FIX 4.2 Protocol for *PossResend* (97) = Y . **Order Cancel Request (35=F)** 'N' (default) indicates a new cancel. 'Y' Indicates an application level unsolicited resend. If *ClOrdID* has not yet been seen, the cancel is treated as normal. If *ClOrdID* already exists, the resent cancel is ignored. **Order Cancel/Replace Request (35=G)** 'Y' Indicates an application level resend. If the *ClOrdID* does not indicate an already pending **`Cancel/Replace`** message, the **`Cancel`** message is treated as normal. If *ClOrdID* does indicate an already pending **`Cancel/Replace`** message, the resent **`Cancel/Replace`** message is ignored. 'N' (default) indicates a new **`Cancel/Replace`** message.

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Standard Message Trailer Message Fields

A standard trailer terminates each message.

Table 15. Standard Message Trailer Message Fields

TAG	FIELD NAME	DESCRIPTION
10	*CheckSum*	Modulo 256 checksum of all characters in message up to and including the delimiter preceding the *CheckSum* field. Three digits with leading zeros if necessary.

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Order Protocol - Member to Cboe Canada

New Order Single Message Fields

Table 16. New Order Single Message Fields

TAG	FIELD NAME	REQ'D	DESCRIPTION
35	Standard Message Header	Y	*MsgType*=D
1	*Account*	N	Reflected back on Execution Reports associated with this order. Characters in ASCII range 32-126 are allowed. Trailing spaces and characters in excess of 16 will be truncated. Available via Standard FIX Drop on an opt-in basis at the port level. Available by default on Order by Order FIX Drop.
11	*ClOrdId*	Y	Client order identifier
18	*ExecInst*	N	Single value only (with no trailing space) All values not defined below are rejected. **MATCHNow:** IOC Orders (may specify *ExecInst*): N = Trade at any eligible price (Default). b = Trade at any eligible price within the PNBBO. M= Trade at the PNBBO midpoint only. Non-IOC Orders (must specify *ExecInst)*: R = Trade at the near-side PNBBO (Size and price restrictions apply) P = Trade at the far-side PNBBO (*ConditionalFlag* (8002) = 0 or 1 only) (Size and price restrictions apply) p = Trade at the less aggressive of PNBBO midpoint or minimum improvement from the PNBBO. M= Trade at the PNBBO midpoint only. o = Odd Lot Liquidity Provider **Cboe Canada NEO:** N = Trade at any eligible price (Default). M= Trade at the PNBBO midpoint only. R = Trade at the near-side PNBBO p = Trade at the less aggressive of PNBBO midpoint or minimum improvement from the PNBBO. r = Late - used with (59)=7 to enter a Late Limit on Close (LLOC)
21	*HandlInst*	Y*	**(Ignored by MATCHNow)** Handling instruction for the order 1 = Direct Action Order (DAO) 5 = Protect and Cancel 6 = Protect and Reprice
23	*IOIid** (MATCHNow only)	N	Must match the *ClOrdID* (11) of the conditional order being firmed up. Must be present if *ConditionalFlag* (8002) = 1. Rejected unless *ConditionalFlag* (8002)= 1.
38	*OrderQty*	Y	Number of shares for order, 1 to 999,999,999.
40	*OrdType*	Y	1 = Market 2 = Limit 3 = Stop (**rejected by MATCHNow**) 4 = Stop Limit (**rejected by MATCHNow**) P = Pegged (**rejected by MATCHNow**)

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TAG	FIELD NAME	REQ'D	DESCRIPTION
44	*Price*	N	Limit Price. Required for limit orders. If populated for a market order (*OrdType* (40) = 1), the order will be accepted and behave like a market order (i.e. *Price* will be ignored). If applied to a pegged order (*OrdType* = P), indicate the price cap of the order. Order rejected if priced finer than the minimum trading increment for the stock.
54	*Side*	Y	1 = Buy 2 = Sell 5 = Sell Short (client affirms ability to borrow) 8 = Cross (see Cross Orders on page 54) 9 = Cross Short (see Cross Orders on page 54)
55	*Symbol*	Y	Symbol (upper case) (up to 14 characters).
58	*Text*	N	Free format text string. Limit of 64 characters.
59	*TimeInForce*	Y	**MATCHNow:** 0 = DAY 3 = Immediate or cancel (IOC) (only permitted in MATCHNow if *ConditionalFlag* (8002) = 2 or 3) 6 = GTD (same day only, *i.e.* Good Til Time) A = EnC (Execute 'N Cancel: a DAY/GTD order which cancels remainder on a partial fill. Only permitted if *ConditionalFlag* (8002) = 1.) **Cboe Canada NEO:** 0 = DAY 1 = Good Til Cancel (GTC) 2 = At the Open (OPG) 3 = Immediate or cancel (IOC) 4 = Fill or Kill (FOK) 6 = Good Til Date (GTD), Good Til Time (GTT) 7 = At the Close (ATC) R = Regular Hours Only (RHO)
60	*TransactTime*	Y	Time order initiated/released. Required by FIX 4.2, but not used by Cboe Canada.
63	*SettlmntTyp*	N	**(Ignored by MATCHNow)** Required if *ExDestination* (100) = S Indicates the settlement period. 1 = Cash 6 = Future 11= Non-Net
64	*FutSettDate*	N	**(Ignored by MATCHNow)** (Required if *SettlmntTyp* (63) = 6, otherwise ignored.) Specific date of trade settlement in format YYYYMMDD.
99	*StopPx*	N	**(Ignored by MATCHNow)** The trigger price for Stop and Stop Limit orders. Required if *OrdType*(40) = 3 or 4. Stop and Stop Limit orders will only be triggered by NLSP trades.
100	*ExDestination*	Y*	***(Ignored by MATCHNow, required by Cboe Canada NEO)**

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TAG	FIELD NAME	REQ'D	DESCRIPTION
			Indicates which Cboe Canada trading book the order should be directed to. L = NEO-L N = NEO-N D = NEO-D C = Crossing Facility S = SST
110	*MinQty* (MATCHNow only)	N	Minimum fill quantity for orders which only interact with liquidity on the target Cboe Canada market. Multiple consecutive fills may be created such that their combined size satisfies the *MinQty* requirement. When the remaining size on an order is less than the defined *MinQty*, then *MinQty* will be automatically set to the remaining size if the "Resize Minimum Quantities" port attribute is enabled. See FIX Port Attributes on page 64 for details. Ignored if *TrueMinQty* is also specified.
111	*MaxFloor*	N	**(Ignored by MATCHNow)** Portion of *OrderQty* to display. Used for iceberg orders. Iceberg orders are Limit orders (40= 2) having *MaxFloor* specified. This is the displayed quantity. The balance (*OrderQty - MaxFloor*) is the reserve. The displayed quantity of each order at a price level is decremented first. Integer value > 0 and <= *OrderQty*. **Ignored** if order is IOC or *OrderType*(40) = 2 (Limit).
126	*ExpireTime*	N	Specifies the UTC Date/Time timestamp that the order expires. Values may be specified at a microsecond level using format YYYYMMDD-HH:MM:SS.ssssss. Valid timestamps must have today's date and a time within trading hours. *ExpireTime* is optional for *TimeInForce* (59) = A (EnC). **MATCHNow only**: *ExpireTime* is Required for *TimeInForce* (59) = 6 (GTD).
211	*PegDifference*	N	**MATCHNow:** Signed dollar value in increments of $0.005 which is added to the result of peg calculations. If the peg is not on a market increment, the working price will be next available market increment in the signed direction. For example, a primary buy peg of .025 when the market bid is 0.490 will work at 0.520 (bid+0.030) since prices greater than $0.50 are in whole penny increments. Order rejected if *PegDifference* is specified without *ConditionalFlag* (8002) = 0 (Conditional Order) or 1 (firm-up of conditional). **Cboe Canada NEO:** Optional tag for use with Primary Peg orders (*ExecInst* = R). If not sent or sent with a zero value, the order will peg directly to its respective side of the PNBBO. Signed dollar value up to two (three) decimal places is added to result of peg calculation when the peg difference is at or above (below) $0.50. NOTE: Only non-aggressive offsets are supported for visible primary pegs, i.e., pegging at or away from the PNBBO; visible buy orders cannot have a positive value, and visible sell orders cannot have a negative value.

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TAG	FIELD NAME	REQ'D	DESCRIPTION
			Offset entered must be tick-aligned. The system will appropriately round down (up) to the nearest 1 cent tick for buy (sell) peg offsets with half cent offset amounts, if the calculated price is at or above 50 cents. Order rejected if *PegDifference* is specified without *ExecInst* (18) = R (Primary Peg).
432	*ExpireDate*	N	**(Ignored by MATCHNow)** Specifies the date on which a GTD order will expire using format YYYYMMDD. Orders will expire at the end of the *ExpireDate*. Rejected if the date is before the current date. Required if *TimeInForce* is 6 (GTD) and *ExpireTime* is not specified. Ignored if *TimeInForce* is not 6 (GTD) or if *ExpireTime* is present.
1724	*LEIOrderOrigination*	N	Identify the type of customer sending the order. 5 = Direct access client 6 = Foreign dealer equivalent 7 = Order execution only service
2883	*LEIRoutingArrangementIndicator*	N	Routing arrangement is in place (1 or 0, meaning yes or no, respectively).
6750	*UMIRAccountType*	Y	The capacity for the order. IN= Inventory NC= Non-Client CL= Client ST= Specialist OF= Options Firm Account OT= Options Market Maker BU= Bundled MC= Multi Client
6751	*UMIRUserID*	Y	Identifies the trader responsible for the order (up to 12 alphanumeric). Also known as Trader ID.
6754	*BasketTrade*	N	Identification for order as part of a basket trade: N = No (Default) integer= Basket Number (non-zero integer, 1-5 digits).
6755	*ProgramTrade*	N	Identify this order as part of a program trade (Y or N). **Default** is N.
6757	*Jitney*	N	Broker on whose behalf this order is being submitted if different than the executing broker. (3-digit numeric identifier, with leading zeros when necessary).
6761	*Anonymous*	N	Anonymize this order when trade reporting (Y or N). **Default** is N.
6763	*UMIRRegulationID*	N	Does not affect order handling but is reported to feeds. SS= Significant Shareholder IA= Insider
6767	*BuyAccountType*	N	See Cross Orders on page 54.
6768	*SellAccountType*	N	See Cross Orders on page 54.
6769	*BuyAccountID*	N	See Cross Orders on page 54.
6770	*SellAccountID*	N	See Cross Orders on page 54.
6771	*BuyRegulationID*	N	See Cross Orders on page 54.
6772	*SellRegulationID*	N	See Cross Orders on page 54.
6773	*CrossType*	N	See Cross Orders on page 54.

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TAG	FIELD NAME	REQ'D	DESCRIPTION
6781	*BuyJitney*	N	See Cross Orders on page 54.
6782	*SellJitney*	N	See Cross Orders on page 54.
6783	*NonResident*	N	**(Ignored by MATCHNow)** Identifies an order as Non-Resident (Y/N) **Default** is N.
6791	*ByPass*	N	Identifies an order as eligible to match against the displayed volume only (Y/N). **Default** is N.
6792	*NCIB*	N	Identifies Normal Course Issuer Bid orders (Y/N). **Default** is N.
7713	*NoTradeFeat*	N	Defines how to prevent a self-trade. Self-trade is identified when the buyer and seller have the same Broker ID (*OnBehalfOfCompId)* and the same *NoTradeKey* value. Required if *NoTradeKey* is present. If the older and newer orders in a self-matching trade have conflicting instructions, the instruction of the newer order will be honored. **MATCHNow:** IB= Ignore match (do not trade) EB= Suppress from Tape U = No Self Trade Protection **Cboe Canada NEO:** EB= Suppress from Tape N = Cancel Newest O = Cancel Oldest D = Decrement larger / Cancel Smaller U = No Self Trade Protection
7714	*NoTradeKey*	N	Identifier for self-trade prevention (alphanumeric of up to length 6). Required for self-trade prevention. Required if *NoTradeFeat* has any value other than 'U' (no protection).
7729	*ShortMarkingExempt*	N	Used by regulators to identify directional neutral accounts. Order will reject if *ShortMarkingExempt(7729)*= 0 and *Side*(54) = 5 (sell short). Not required, but if present must be: 0 = SME (Non cross order)
8002	*ConditionalFlag* (MATCHNow only)	N	Identify this order as a conditional order. 0 = Conditional Order (*DAY and GTD only*) 1 = Firm Up of Conditional Order 2 = Eligible to interact with Conditionals (*DAY, GTD, and IOC only*) 3 = (Default). Ineligible to interact with Conditionals
8025	*LEICustomerAccount*	N	Account number for clients not eligible to obtain an LEI. All characters in ASCII range 32-126 are allowed, excluding comma, semicolon, and pipe.
8026	*LEIAlgorithmID*	N	Unique identifier for the end-client orders automatically generated on a predetermined basis. All characters in ASCII range 32-126 are allowed, excluding comma, semicolon, and pipe.
8027	*CustomerLEI*	N	LEI for clients eligible to obtain an LEI including LEI of the foreign dealer equivalent (Encryption Required). (base64 characters: alphanumeric, including plus sign, forward slash, and equals)
8028	*BrokerLEI*	N	Non-PO IIROC Dealer Member (Correspondent Broker)

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TAG	FIELD NAME	REQ'D	DESCRIPTION
			(alphanumeric of length 2 to 20).
9303	*RoutingInst*	N	**(Ignored by MATCHNow)** P = Post Only (cancel) p = Post Only (reprice)
9479	*DisplayIndicator*	N	**(Ignored by MATCHNow)** V = visible I = invisible
20005	*MatchingStateParticipation*	N	**(Ignored unlessExDestination = D)** Indicates which order flow type a Passive Dark order elects to interact with. Passive Dark orders have a *TimeInForce* of Day or RHO. Active Orders have a *TimeInForce* of IOC or FOK. 1 = Active orders only 2 = Passive orders only 3 = Both active and passive orders (**Default**)
20201	*BuyCustomerAccount*	N	See Cross Orders on page 54.
20202	*SellCustomerAccount*	N	See Cross Orders on page 54.
20203	*BuyAlgorithmID*	N	See Cross Orders on page 54.
20204	*SellAlgorithmID*	N	See Cross Orders on page 54.
20205	*BuyCustomerLEI*	N	See Cross Orders on page 54.
20206	*SellCustomerLEI*	N	See Cross Orders on page 54.
20207	*BuyBrokerLEI*	N	See Cross Orders on page 54.
20208	*SellBrokerLEI*	N	See Cross Orders on page 54.
20209	*BuyOrderOrigination*	N	See Cross Orders on page 54.
20210	*SellOrderOrigination*	N	See Cross Orders on page 54.
20211	*BuyRoutingArrangement*	N	See Cross Orders on page 54.
20212	*SellRoutingArrangement*	N	See Cross Orders on page 54.
22009	*TrueMinQty*	N	All fills must be at least of size *TrueMinQty*. When the remaining size on an order is less than the defined *TrueMinQty*, then *TrueMinQty* will be automatically set to the remaining size if the "Resize Minimum Quantities" port attribute is enabled. Order rejected if *TrueMinQty* is an odd or mixed-lot quantity. **Default** is zero.
	Standard Message Trailer	Y	

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Order Cancel Request Message Fields

Request to cancel an order.

Table 17. Order Cancel Request Message Fields

TAG	FIELD NAME	REQ'D	DESCRIPTION
35	Standard Message Header	Y	*MsgType*=F
11	*ClOrdID*	Y	Day-unique cancel ID chosen by Member. 20 characters or less. Characters in ASCII range 33-126 are allowed, except for comma, semicolon, and pipe. Duplicate order *ClOrdID*s will be rejected (or ignored if *PossResend*= Y).
37	*OrderID*	N	*OrderID* supplied by Cboe Canada on the order acknowledgement. *ClOrdID* must be sent if *OrderID* is not.
41	*OrigClOrdID*	N	*ClOrdID* of the order to cancel. *OrderID* must be sent if *ClOrdID* is not.
60	*TransactTime*	N	Time cancel initiated/released. Required by FIX 4.2 but not used by Cboe Canada.
1724	*LEIOrderOrigination*	N	Identify the type of customer sending the order. 5 = Direct access client 6 = Foreign dealer equivalent 7 = Order execution only service
2883	*LEIRoutingArrangementIndicator*	N	Routing arrangement is in place (1 or 0, meaning yes or no, respectively).
8025	*LEICustomerAccount*	N	Account number for clients not eligible to obtain an LEI. All characters in ASCII range 32-126 are allowed, excluding comma, semicolon, and pipe.
8026	*LEIAlgorithmID*	N	Unique identifier for the end-client orders automatically generated on a predetermined basis. All characters in ASCII range 32-126 are allowed, excluding comma, semicolon, and pipe.
8027	*CustomerLEI*	N	LEI for clients eligible to obtain an LEI including LEI of the foreign dealer equivalent (Encryption Required). (base64 characters: alphanumeric, including plus sign, forward slash, and equals)
8028	*BrokerLEI*	N	Non-PO IIROC Dealer Member (Correspondent Broker) (alphanumeric of length 2 to 20)
	Standard Message Trailer	Y	

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Order Cancel/Replace Request Messages

Request to modify an order.

Changes in *OrderQty* result in an adjustment of the current order's *OrderQty*. The new *OrderQty* does not directly replace the current order's *LeavesQty*. Rather, a delta is computed from the current *OrderQty* and the replacement *OrderQty*. This delta is then applied to the current *LeavesQty*. If the resulting *LeavesQty* is less than or equal to zero, the order is cancelled. This results in safer behavior when the modification request overlaps partial fills for the current order, leaving the Member in total control of the share exposure of the order.

The **`Cancel/Replace`** message will only be accepted if the original order can be successfully removed. Requests which cannot be processed will be rejected using the **`Cancel Reject`** message. The **`Cancel Reject`** message will provide the *ClOrdID* and *OrigClOrdID* values which were specified on the **`Cancel/Replace`** message for identification.

A **`Cancel/Replace`** message should not be issued until the ack for the previous **`Cancel/Replace`** message has been received for that order (or the acknowledgement for the first **`Cancel/Replace Request`** message). The FIX handler will reject a new **`Cancel/Replace`** message if it has not seen the prior **`Cancel/Replace`** message from the Matching Engine.

A maximum of 1,679,615 **`Cancel/Replace`** message requests may be made to a single order each trading day. Once the 1,679,615th modification is made, the next user-generated message on the order should be an **`Order Cancel`** message request.

The presence of any of *Program Trade*, *Jitney*, *UMIRRegulationID*, or *Short Sale Exempt* will cause the message to be rejected.

Modifies will lose time priority of the order unless the modification involves only a decrease *OrderQty*, a change to *MaxFloor*, and/or a change in *StopPx*.

Modifications to GTT/GTD orders which do not include an *ExpireTime* or *ExpireDate* tag will preserve their existing expiration.

Other fields (including *ExecInst*) **will be ignored**, and the value from the original order will be re-used.

Table 18. Modifiable Field Summary

TAG	FIELD NAME	MATCHNOW	CBOE CANADA NEO
1	*Account*	No	Yes
38	*Quantity*	Yes	Yes
40	*OrdType*	Yes	No
44	*Price*	Yes	Yes
59	*TIF*	No	Yes
99	*StopPx*	No	Yes
111	*MaxFloor*	No	Yes
126	*ExpireTime*	No	Yes
432	*ExpireDate*	No	Yes

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TAG	FIELD NAME	MATCHNOW	CBOE CANADA NEO
1724	*LEIOrderOrigination*	Yes	Yes
2883	*LEIRoutingArrangementIndicator*	Yes	Yes
6750	*UMIRAccountType*	Yes	Yes
8025	*LEICustomerAccount*	Yes	Yes
8026	*LEIAlgorithmID*	Yes	Yes
8027	*CustomerLEI*	Yes	Yes
8028	*BrokerLEI*	Yes	Yes
22009	*TrueMinQty*	No	Yes

Table 19. FIX Message Fields

TAG	FIELD NAME	REQ'D	DESCRIPTION
35	Standard Message Header	Y	*MsgType* = G
1	*Account*	N	**MATCHNow Only:** **Ignored** - value preserved from original order. **Cboe Canada NEO:** Reflected back on Execution Reports associated with this order. 16 or fewer characters in the ASCII range.
11	*ClOrdId*	Y	Day-unique ID chosen by Member. Duplicate order *ClOrdID*s will be rejected[1] (or ignored if *PossResend* = Y).
18	*ExecInst*	N	**Ignored** - value preserved from original order
21	*HandlInst*	N	**Ignored** - value preserved from original order
23	*IOIid*	N	**Ignored** - value preserved from original order
37	*OrderID*	N	*OrderID* supplied by Cboe Canada on the order acknowledgement. In the case of multiple changes to a single order, this will be the *OrderID* from the most recent acknowledgement.
38	*OrderQty*	N	Number of shares for order. This will modify the *OrderQty* of the current order; it does **not** directly set the remaining quantity.
40	*OrdType*	N	**MATCHNow Only:** 1 = Market 2 = Limit Defaults to original order if not sent.May replace Limit with Market and vice versa. **Cboe Canada NEO:** **Ignored** - value preserved from original order
41	*OrigClOrdID*	N	*ClOrdID* of the order to replace. Required if *OrderID* (37) is not present, otherwise ignored. In the case of multiple changes to a single order, this will be the *ClOrdID* of the **most recent** accepted change.
44	*Price*	N	Limit Price. Order rejected if priced finer than the minimum trading increment for the symbol.
54	*Side*	N	**Ignored** - value preserved from original order
55	*Symbol*	N	**Ignored** - value preserved from original order
58	*Text*	N	**Ignored** - value preserved from original order
59	*TimeInForce*	N	**(Ignored by MATCHNow)** Modifiable to any of the following *TimeInForce* values if the current *TimeInForce* value is also any of the following:

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TAG	FIELD NAME	REQ'D	DESCRIPTION
			0 = Day 1 = Good Til Cancel (GTC) 6 = Good Til Date (GTD), Good Til Time (GTT) R = Regular Hours Only (RHO)
60	*TransactTime*	Y	Time Cancel/Replace initiated/released.
63	*SettlmntTyp*	N	**Ignored** - value preserved from original order
64	*FutSettDate*	N	**Ignored** - value preserved from original order
99	*StopPx*	N	**(Ignored by MATCHNow)** The trigger price for Stop and Stop Limit orders. Ignored unless *OrdType*(40) or the original order is equal to 3 or 4. If not present, the value preserved from original order.
100	*ExDestination*	N	**Ignored** - value preserved from original order
111	*MaxFloor*	N	**(Ignored by MATCHNow)** New size to expose.
126	*ExpireTime*	N	**(Ignored by MATCHNow)** Specifies the time of day when a GTT (GTD) order will expire. Rejects if the time is not for the current date. Ignored if *TimeInForce* is not GTD.
211	*PegDifference*	N	**Ignored** - value preserved from original order.
432	*ExpireDate*	N	**(Ignored by MATCHNow)** Specifies the date on which a GTD order will expire. Ignored if *TimeInForce* is not GTD. Ignored if *ExpireTime* is also present.
1724	*LEIOrderOrigination*	N	Identify the type of customer sending the order. 5 = Direct access client 6 = Foreign dealer equivalent 7 = Order execution only service
2883	*LEIRoutingArrangementIndicator*	N	Routing arrangement is in place (1 or 0, meaning yes or no, respectively).
6750	*UMIRAccountType*	N	The capacity for the order. IN= Inventory NC= Non-Client CL= Client ST= Specialist OF= Options Firm Account OT= Options Market Maker BU= Bundled MC= Multi Client
6751	*UMIRUserID*	N	**Ignored** - value preserved from original order
6754	*BasketTrade*	N	**Ignored** - value preserved from original order
6755	*Program Trade*	N	**Reject** if present.
6757	*Jitney*	N	**Reject** if present.
6761	*Anonymous*	N	**Ignored** - value preserved from original order
6763	*UMIRRegulationID*	N	**Reject** if present.
6783	*NonResident*	N	**(Ignored by MATCHNow)** **Ignored** - value preserved from original order
6791	*ByPass*	N	**Ignored** - value preserved from original order
6792	*NCIB*	N	**Ignored** - value preserved from original order
7713	*NoTradeFeat*	N	**Ignored** - value preserved from original order
7714	*NoTradeKey*	N	**Ignored** - value preserved from original order
7729	*ShortMarkingExempt*	N	**Reject** if present.

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TAG	FIELD NAME	REQ'D	DESCRIPTION
8002	*ConditionalFlag*	N	**Ignored** - value preserved from original order
8025	*LEICustomerAccount*	N	Account number for clients not eligible to obtain an LEI. All characters in ASCII range 32-126 are allowed, excluding comma, semicolon, and pipe.
8026	*LEIAlgorithmID*	N	Unique identifier for the end-client orders automatically generated on a predetermined basis. All characters in ASCII range 32-126 are allowed, excluding comma, semicolon, and pipe.
8027	*CustomerLEI*	N	LEI for clients eligible to obtain an LEI including LEI of the foreign dealer equivalent (Encryption Required). (base64 characters: alphanumeric, including plus sign, forward slash, and equals)
8028	*BrokerLEI*	N	Non-PO IIROC Dealer Member (Correspondent Broker) (alphanumeric of length 2 to 20).
9619	*CancelOrigOnReject*	N	N = Leave original order alone. Y = Cancel original order if replacement fails (an unsolicited Cancel report will be sent for original order in this case). Order will not be cancelled if the symbol is in a "no cancellations" state. Default is configurable per port (N if not configured).
20005	*MatchingStateParticipation*	N	**Ignored** - value preserved from original order
22009	*TrueMinQty*	N	**(Ignored by MATCHNow)** Min quantity for a single fill. **Modifiable** if present on the original order, otherwise rejected. Cannot be modified to 0.
	Standard Message Trailer	Y	

[1] Cancel/Replace messages that merely reduce *OrderQty* may be overlapped if the existing *ClOrdID* is re-used. This is the only case where re-use of the existing *ClOrdID* is allowed.

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Order Protocol - Cboe Canada to Member

Execution Report Fields

The Execution Report Fields table contains all possible fields that are sent in Execution Reports. Not all fields are sent on each Execution Report. Additional or unexpected FIX fields on all FIX messages should be ignored. Note that fields described as "copied from order" may include values applied via defaults or port settings.

Table 20. Execution Report Fields

TAG	FIELD NAME	REQ'D	DESCRIPTION
35	Standard Message Header	Y	*MsgType*= 8
1	*Account*	N	Copied from order. For Order by Order FIX Drop, all execution information will be sent.
6	*AvgPx*	Y	Average price of executions for this order weighted by trade size. Zero if *CumQty* is zero.
11	*ClOrdID*	Y	ClOrdID of the order being accepted, executed, or rejected. -or- ClOrdID of the cancel or replace request. -or- ClOrdID of the order subject to unsolicited cancel (OrigClOrdID will not be present).
14	*CumQty*	Y	Cumulative quantity of shares executed for this order.
15	*Currency*	N	The currency of the symbol that has been traded. USD=United States Dollar CAD=Canadian Dollar Present if *ExecType* (150) = 1 (partial filled) or 2 (filled).
17	*ExecID*	Y	Day-unique id of execution message.
18	*ExecInst*	N	Copied from order.
20	*ExecTransType*	Y	0=New 3=Status. Used to provide order restatements (150=D). 5=Auto execution
21	*HandInst*	N	Copied from order.
31	*LastPx*	N	Price of this fill (zero for non-fills).
32	*LastShares*	N	Quantity of shares traded on this fill (zero for non-fills).
37	*OrderID*	Y	Day-unique order identifier (supplied by Cboe Canada).
38	*OrderQty*	Y	Copied from order. On restatements (150=D) the *OrderQty* may be updated.
39	*OrdStatus*	Y	State of order. 0=New 1=Partially Filled 2=Filled 3=DoneForDay 4=Canceled 5=Replaced 6=Pending Cancel 8=Rejected E=Pending Replace

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TAG	FIELD NAME	REQ'D	DESCRIPTION
			For FIX Drop, only '1' or '2' will be sent and will always equal *ExecType* (150). For Order by Order FIX Drop, all execution information will be sent.
40	*OrdType*	N	Copied from order.
41	*OrigClOrdID*	N	*ClOrdID* of the order being cancelled or replaced (for a solicited **`Cancel`** or **`Cancel/Replace`** message, otherwise not present).
44	*Price*	N	Copied from order.
54	*Side*	Y	Copied from order.
55	*Symbol*	Y	Copied from order.
58	*Text*	N	If present, indicates reason for reject or cancel. Format is one letter reason code followed by colon and space followed by free form text message. (e.g. 'N : No Liquidity at price'). The specific text the system delivers may vary from the text listed below, to provide clarification of the reject reason. **Reason codes are one of the following:** A=Admin (includes many formatting & tag value errors) D=Duplicate identifier (e.g., *ClOrdID*) H=Halted I=Incorrect data center J=Too late to cancel K=Order rate threshold exceeded M=Order size exceeded N=Not Executed O=*ClOrdID* doesn't match a known order U=User requested X=Order expired Y=Symbol not supported Z=Unforeseen reason j=Order is below min required quantity u=Circuit breaker threshold breach - LimitUpDown y=Order received by Cboe Canada during replay
59	*TimeInForce*	N	Copied from order.
60	*TransactTime*	Y	GMT date-time that transaction occurred.
63	*SettlmtTyp*	N	Copied from order.
64	*FutSettDate*	N	Copied from order.
75	*TradeDate*	N	Indicates the date of the trade referenced in this message in YYYYMMDD format. Absence of this field indicates that the current date is the trade date. This field is used to indicate a T+1 trade for NEO Connect trades.
76	*ExecBroker* (MATCHNow only)	N	Identifies the Introducing Broker if applicable. This tag is present only on drop feeds, and only for conditional sponsored broker orders and trades. When an Introducing Broker is not used, it will identify the Executing Broker.
99	*StopPx*	N	Copied from order.
100	*ExDestination*	N	Copied from order.
103	*OrdRejReason*	N	Only used when *ExecType* is Rejected (8). 0=Exchange option 1=Unknown symbol

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TAG	FIELD NAME	REQ'D	DESCRIPTION
			2=Exchange closed 3=Order exceeds limit 5=Unknown order 6=Duplicate order 8=Stale order
109	*ClientID* (MATCHNow only)	N	Indicates client identifier (up to 16 characters). This tag is present when *ExecType* (150)= 2 (filled) for Dark Pool originated liquidity, otherwise not present if *ExecType* (150) = 1 (partially filled) or 2 (filled).
110	*MinQty*	N	Copied from order.
111	*MaxFloor*	N	Copied from order.
126	*ExpireTime*	N	Copied from order if *TimeInForce* is GTD (6) or EnC (A).
150	*ExecType*	Y	Reason for this Execution Report: 0=New (acknowledgement of new order) 1=Partial Fill 2=Fill 3=DoneForDay 4=Canceled 5=Replaced 8=Rejected D=Restated For Standard FIX Drop, only 1 or 2 will be sent and will always equal *OrdStatus* (39). For Order by Order FIX Drop, all execution information will be sent. Refer to Independent Software Vendor Configuration on page 60.
151	*LeavesQty*	Y	Quantity of shares still open for further execution. Will be zero if order is dead, otherwise will be *OrderQty* (38) - *CumQty* (14). **Note:** It is possible for *LeavesQty* to be zero when *ExecType* (150) = 5 indicating that the order is dead.
198	*SecondaryOrderID*	N	Present on a Self-Trade Prevention triggered cancel/restatement, on a reserve reload restatement, and on a Cancel/Replace acknowledgement that increases the size of a reserve order. Contra side Cboe *OrderID* of prevented self-trade, or Cboe PITCH *OrderID* for displayed portion of reserve order after reload, or Cboe PITCH *OrderID* for the displayed portion of a reserve order after a Cancel/Replace increased the size of the reserve order.
375	*ContraBroker*	N	The Broker ID (3-digit identifier) of the contra party on this fill. This tag is present if (and only if) *OrdStatus* is 1 (partially filled) or 2 (filled). *ContraBroker* will be '001' if the contra party was marked as anonymous. Note: this is a standalone tag and is not used in conjunction with the *NoContraBrokers* (382) repeating group.
378	*ExecRestatementReason*	N	Only present when 150=D or 150=3. 1=GTC and GTD restatement 3=Repricing (ex. OPR, MTL) 4=Iceberg display size refreshed 5=Partial Reduce due self- trade prevention result based on *NoTradeFeat* (7713)

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TAG	FIELD NAME	REQ'D	DESCRIPTION
			18=Triggered stop order
432	*ExpireDate*	N	Copied from order.
851	*LastLiquidityInd*	N	Present for fills (*ExecType* (150) = 1 (partially filled), 2 (filled). 1=Trade added liquidity 2=Trade removed liquidity 4=Auction Trade 8=Conditional Trade (**MATCHNow only**)
1138	*DisplayQty*	N	The quantity of the order displayed in the order book.
1724	*LEIOrderOrigination*	N	Copied from order.
2883	*LEIRoutingArrangementIndicator*	N	Copied from order.
6750	*UMIRAccountType*	N	Copied from order.
6751	*UMIRUserID*	N	Copied from order.
6754	*BasketTrade*	N	Copied from order.
6755	*ProgramTrade*	N	Copied from order.
6757	*Jitney*	N	Copied from order.
6761	*Anonymous*	N	Copied from order.
6763	*UMIRRegulationID*	N	Copied from order.
6776	*PrincipalTrade*	N	Present on executions where Broker executed a Principal trade. If present, value is Y.
6777	*WashTradeIndicator*	N	Present on executions where Broker executed a wash trade. If present, value is Y.
6783	*NonResident*	N	**(Ignored by MATCHNow)** Copied from order.
6791	*ByPass*	N	Copied from order.
6792	*NCIB*	N	Copied from order.
7713	*NoTradeFeat*	N	Copied from order if ExecType(150) = 0 (new), '1' (partially filled), or '2' (filled).
7714	*NoTradeKey*	N	Copied from order if ExecType(150) = 0 (new), '1' (partially filled), or '2' (filled).
7729	*ShortMarkingExempt*	N	Copied from order.
7732	*MatchingPriority*	N	The type of priority used to match an order in a trade. May only be present if 150 = 1 or 2 (partial fill or fill). 1=Broker Preferencing 100=Market Maker Preferential Trading (note: only applies to market makers).
7733	*SelfTrade*	N	Present on executions where Broker executed a self-trade using *NoTradeFeat* (7713) = EB (suppress from tape). If present, value is 'Y'. This tag is optionally present if *ExecType* (150) = 1 (partially filled) or '2' (filled).
8002	*ConditionalFlag*	N	Copied from order.
8005	*ConditionalStatus*	N	Present on a conditional order's firm-up invitation. 5=Firm order request message
8025	*LEICustomerAccount*	N	Copied from order.
8026	*LEIAlgorithmID*	N	Copied from order.
8027	*CustomerLEI*	N	Copied from order.
8028	*BrokerLEI*	N	Copied from order.

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TAG	FIELD NAME	REQ'D	DESCRIPTION
9617	*ModifySequence*	N	FIX Drop only. Base 36 number of times order has been replaced.
9688	*OrigCompID*	N	FIX Drop only. *TargetCompID* of original FIX exec report. FIX Drop port must be configured to send this optional field.
9689	*OrigSubID*	N	FIX Drop only. *TargetSubID* of original FIX exec report. FIX Drop port must be configured to send this optional field.
9882	*FeeCode*	N	Specific fee code associated with execution. See the Fee Schedule of the respective market for possible values. Fee codes may be 1 or 2 alphanumeric characters.
20005	*MatchingStateParticipation*	N	Copied from order.
22009	*TrueMinQty*	N	Copied from order.
	Standard Message Trailer	Y	

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Cancel Reject Message Fields

Rejects a **`Cancel`** or **`Cancel/Replace`** message request.

When aCancel/Replacemessage is rejected, by default the original order is left alive. ACancel Reject message should not be used as a sign that the original order has been cancelled. Even if the CancelOrigOnReject =Y option is being used a separate unsolicited cancel will be sent to close out the original order.

Table 21. Cancel Reject Message Fields

TAG	FIELD NAME	REQ'D	DESCRIPTION
35	Standard Message Header	Y	*MsgType*=9
11	*ClOrdID*	Y	*ClOrdID* from the **`Cancel`** message or **`Cancel/Replace`** message request.
37	*OrderID*	Y	*OrderID* of order that failed to be cancelled or replaced. Represents the originating/initial order in an order chain. NONE if *CxlRejReason* is Unknown (1).
39	*OrdStatus*	Y	Order Status of order that failed to be cancelled or replaced.
41	*OrigClOrdID*	Y	*ClOrdID* of the order that failed to be cancelled or replaced.
58	*Text*	N	Free form text message.
102	*CxlRejReason*	N	0 = Too late to cancel 1 = Unknown order 3 = Already pending cancel or pending replace
434	*CxlRejResponseTo*	Y	This reject is in response to a: 1 = Cancel 2 = Cancel/Replace
	Standard Message Trailer	Y	

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Trade Cancel/Correct Message Fields

Sends a **`Trade Cancel/Correct`** message for a trade break.

Trade Cancel/Correctmessage (UCC) is an optional message that is enabled by default at the port level. The price or size of a trade may be corrected, all other details remain the same. Trade cancels and corrections do not alter live order state.

Table 22. Trade Cancel/Correct Message Fields

TAG	FIELD NAME	REQ'D	DESCRIPTION
35	Standard Message Header	Y	*MsgType*= UCC
1	*Account* (Cboe Canada NEO only)	N	Trading account specified by the trader.
11	*ClOrdID*	Y	*ClOrdID* of the original trade being cancelled/corrected.
17	*ExecID*	Y	Day-unique id of execution message.
19	*ExecRefID*	Y	Refers to the *ExecID* of the message being cancelled or corrected.
20	*ExecTransType*	Y	1 = Trade Cancel 2 = Trade Correct
31	*LastPx*	Y	Price on the original trade being cancelled/corrected.
32	*LastShares*	Y	Quantity of shares on the original trade being cancelled/corrected.
37	*OrderID*	Y	*OrderID* of the original trade being cancelled/corrected.
42	*OrigTime*	Y	GMT date-time of original trade.
54	*Side*	Y	Copied from original trade being cancelled/corrected.
55	*Symbol*	Y	Copied from original trade being cancelled/corrected (available in FIX Drop).
60	*TransactTime*	Y	GMT date-time of cancel/correct.
65	*SymbolSfx* (MATCHNow only)	Y	Copied from original trade being cancelled/corrected (available in FIX Drop).
851	*LastLiquidityInd*	Y	Copied from original trade being cancelled/corrected.
6655	*CorrectedSize* (Cboe Canada NEO only)	N	Only for Trade Corrects. Corrected size.
6751	*UMIRUserID* (Cboe Canada NEO only)	Y	Identifies the trader responsible for the order being cancelled/corrected (up to 12 alphanumeric). Also known as Trader ID.
9620	*CorrectedPrice*	N	Only for Trade Corrects. Corrected price.
9882	*FeeCode*	N	Only for Trade Corrects. Corrected fee code.
25021	*FixCorrectedFutSettlDate* (Cboe Canada NEO only)	N	Only for Trade Corrects. Corrected *FutSettDate*.
25022	*FixCorrectedSettlementType* (Cboe Canada NEO only)	N	Only for Trade Corrects. Corrected *SettlementType*.
	Standard Msg Trailer	Y	

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Special Order Types and Features

Special Order Types and Features provides clarifying information that will enable Members to successfully enter orders on MATCHNow.

This is not meant to be an exhaustive list of all FIX tags that accompany that order type; rather, this identifies the key tags and provides additional information regarding certain order types and features. Please refer to New Orders on page 30 for field description details.

Conditional Orders (MATCHNow Only)

Cboe Canada supports conditional orders via a two-step process consisting of conditional order matching (uncommitted orders) and firm order matching (committed orders). A conditional order is entered and later firmed up for execution via the Conditional Orders process. Note that a regular liquidity-providing order can elect to interact with conditional orders.

Conditional Order

- Conditional orders sent to Cboe Canada represent uncommitted quantity and order parameters that the participant is potentially willing to trade. Cboe Canada will only use conditional orders to match participants together and it will only execute trades once firm orders are received.
- Conditional orders must be of large size: they must be **either** at least 51 board lots and have at least $30,000 notional value or have at least $100,000 notional value (in the currency of the symbol being traded). Any orders submitted below this threshold will be rejected. Liquidity-providing orders must also meet these same size and notional value checks to interact with the conditional orders, if so elected. Once the remaining quantity of the Liquidity providing order falls below the defined thresholds it will no longer be eligible to trade conditionally.

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Invitation

- When Cboe Canada identifies a contra to the conditional order, a request is sent inviting the subscriber to firm up. The invitation will always be for the full conditional order quantity sent to Cboe Canada so that no size information about the contra-party will be disclosed.
- The firm-up request comes in the form of an unsolicited FIX status message of the original conditional order with tags as described below. The subscriber is expected to send a firm-up order on receipt of the invitation, within 1 second.
  - *OrdStatus* (39) = 0 (New)
  - *ExecTransType* (20) = 3 (Status)
  - *ExecType* (150) = D (Restated)
  - *ConditionalStatus* (8005) = 5 (Invited)

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Response (firm-up order)

- The participant may respond to the invitation by sending a firm order to Cboe Canada (i.e. an order having tag *IOIid* (23) populated with the *ClOrdID* (11) value of the conditional and *ConditionalFlag* (8002) = 1).
- The participant should never send an order with an *IOIid* (23) unless responding to an invitation. Cboe Canada will reject any firm-up orders that are not in response to a live invitation.

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Response (rejection)

- The participant may actively reject an invitation by sending a cancellation request to Cboe Canada for the conditional order.

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Result (firm-up traded)

- If traded, Cboe Canada will send a cancellation of the conditional order followed by an execution report of the firm-up order with a fill or partial fill order status.
- The subscriber may send a new conditional order after the fill to represent the remaining quantity in the Cboe Canada conditional venue.
- All trades will be priced at or inside the current Protected National Best Bid/Offer (PNBBO).
- All firm-up trades will be broker-preferenced.
- If partially traded, the firm-up order will remain open and tradable with future interactions.

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Result (no trade of firm-up)

- Orders may not trade due to various reasons including limit prices, min quantities, or the contra-side not responding in time to their own invitation.
- If not traded (or only partially traded), the firm-up order will remain open and tradable with future interactions.
- Firm-up orders can be cancelled at any time. Expiry times are also supported and can be used at any time.

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Cross Orders

Pre-arranged, two-sided trades can be reported and printed via Cboe Canada. To do so, a New Order Single message is used (35 = D) with a *Side* (54) value of 8 (Cross) or 9 (Cross Short). A different set of tag are then used to describe the buyer and seller details.

- Since the New Order Single is being used to represent both the buyer and the seller, Cboe Canada will respond to the order with a total of four Execution Report messages:
  - Execution Report #1 (ack the buyer): 150=0 (new), 54=1(buy)
  - Execution Report #2 (ack the seller): 150=0 (new), 54=2(sell) or 54=5(sell short)
  - Execution Report #3 (fill the buyer): 150=2 (filled), 54=1(buy)
  - Execution Report #4 (fill the seller): 150=2 (filled), 54=2(sell) or 54=5(sell short)
- If the cross is rejected, a single reject message will be sent (35=3, or 35=D and 40=8).
- The *ClOrdID* provided on the New Order Single will be used on all four execution reports.
- There will be 2 OrderID identifiers assigned: one for the buyer (on messages #1 and #3) and one for the seller (on messages #2 and #4).
- The execution reports will, aside from the presence of fields such as *CrossType*, be presented using the regular fields of an execution report, as defined in Cross Order Execution Report on page 57.

Cross Order Entry

All fields not listed are ignored for a cross order entry (i.e. New Order Single having Side (54) = 8 or 9).

Table 23. Cross Order Entry

TAG	FIELD NAME	REQ'D	DESCRIPTION
35	Standard Message Header	Y	*MsgType*=D
11	*ClOrdId*	Y	Day-unique ID chosen by client. Limited to 18 characters for a Cross Orders.
38	*OrderQty*	Y	Number of shares for trade, 1 to 999,999,999.
44	*Price*	Y	Trade Price.
54	*Side*	Y	8=Cross 9=Cross Short
55	*Symbol*	Y	Symbol (upper case) (up to 10 characters).
60	*TransactTime*	Y	Time order initiated/released. Required by FIX 4.2, but not used by Cboe Canada.
63	*SettlmntTyp*	N	Indicates the settlement period. 1=Cash 6=Future 11=Non-Net
64	*FutSettDate*	N	(Required if *SettlmntTyp* (63)=6, otherwise ignored.) Specific date of trade settlement in format YYYYMMDD.
100	*ExDestination*	Y	Must be: C=Cboe Canada Crossing Market

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TAG	FIELD NAME	REQ'D	DESCRIPTION
6751	*UMIRUserID*	Y	Identifies the trader responsible for the order (up to 12 alphanumeric).
6754	*BasketTrade*	N	Identification for order as part of a basket trade: N= No (Default) integer= Basket Number (non-zero integer, 1-5 digits)
6755	*ProgramTrade*	N	Identify this order as part of a program trade (Y or N). **Default** is N.
6761	*Anonymous*	N	Anonymize the buyer and seller of this order when trade reporting (Y or N). **Default** is N.
6767	*BuyAccountType*	Y	See New Order Single Message Fields on page 30, *UMIRAccountType*
6768	*SellAccountType*	Y	See New Order Single Message Fields on page 30, *UMIRAccountType*
6769	*BuyAccountID*	N	See New Order Single Message Fields on page 30, *Account*
6770	*SellAccountID*	N	See New Order Single Message Fields on page 30, *Account*
6771	*BuyRegulationID*	N	See New Order Single Message Fields on page 30, *UMIRRegulationID*
6772	*SellRegulationID*	N	See New Order Single Message Fields on page 30, *UMIRRegulationID*
6773	*CrossType*	N	Identifies the type of cross being printed. B=Basis C=Contingent D=Derivative I=Internal N= Net asset value (NAV)(effective 01/13/26) R=Regular (**Default**) V=VWAP
6781	*BuyJitney*	N	See New Order Single Message Fields on page 30, *Jitney*
6782	*SellJitney*	N	See New Order Single Message Fields on page 30, *Jitney*
6783	*NonResident*	N	**(Ignored by MATCHNow)** Identifies an order as Non-Resident (Y/N) **Default** is N.
6791	*ByPass*	N	Identifies an order as eligible to match against the displayed volume only (Y/N). **Default** is N.
6792	*NCIB*	N	Identifies Normal Course Issuer Bid orders (Y/N). **Default** is N.
7729	*ShortMarkingExempt*	N	Used by regulators to identify directional neutral accounts. Directional neutral account Buy: 54=1, 7729=1 (or 3) Directional neutral account Sell: 54=2, 7729=2 (or 3) 1=Buy Cross SME 2=Sell Cross SME 3=Both Buy and Sell Cross SME
20201	*BuyCustomerAccount*	N	See New Order Single Message Fields on page 30, *LEICustomerAccount*
20202	*SellCustomerAccount*	N	See New Order Single Message Fields on page 30, *LEICustomerAccount*
20203	*BuyAlgorithmID*	N	See New Order Single Message Fields on page 30, LEI*AlgorithmID*
20204	*SellAlgorithmID*	N	See New Order Single Message Fields on page 30, LEI*AlgorithmID*
20205	*BuyCustomerLEI*	N	See New Order Single Message Fields on page 30, *CustomerLEI*
20206	*SellCustomerLEI*	N	See New Order Single Message Fields on page 30, *CustomerLEI*
20207	*BuyBrokerLEI*	N	See New Order Single Message Fields on page 30, *BrokerLEI*
20208	*SellBrokerLEI*	N	See New Order Single Message Fields on page 30, *BrokerLEI*
20209	*BuyOrderOrigination*	N	See New Order Single Message Fields on page 30, *LEIOrderOrigination*

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TAG	FIELD NAME	REQ'D	DESCRIPTION
20210	*SellOrderOrigination*	N	See New Order Single Message Fields on page 30, *LEIOrderOrigination*
20211	*BuyRoutingArrangement*	N	See New Order Single Message Fields on page 30, *LEIRoutingArrangementIndicator*
20212	*SellRoutingArrangement*	N	See New Order Single Message Fields on page 30, *LEIRoutingArrangementIndicator*
	Standard Message Trailer	Y	

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Cross Order Execution Report

The Cross Order Execution Report table contains all possible fields sent in Execution Reports. Not all fields are sent on each Execution Report. Additional or unexpected FIX fields on all FIX messages should be ignored. Fields described as "copied from order" may include values applied via defaults or port settings.

Table 24. Cross Order Execution Report

TAG	FIELD NAME	REQ'D	DESCRIPTION
35	Standard Message Header	Y	*MsgType* = 8
20	*ExecTransType*	Y	0=New
1	*Account*	N	Copied from order (*BuyAccountID* or *SellAccountID*)
6	*AvgPx*	Y	Average price of executions for this order weighted by trade size. Zero if *CumQty* is zero.
11	*ClOrdID*	Y	ClOrdID of the order being accepted or executed.
14	*CumQty*	Y	Cumulative quantity of shares executed for this order.
15	*Currency*	Y	The currency of the symbol that has been traded. USD=United States Dollar CAD=Canadian Dollar Present if *ExecType* (150) = 2 (filled).
17	*ExecID*	Y	Day-unique id of execution message.
20	*ExecTransType*	Y	0=New
31	*LastPx*	Y	Price of this fill (zero for non-fills).
32	*LastShares*	Y	Quantity of shares traded on this fill (zero for non-fills).
37	*OrderID*	Y	*OrderID* (supplied by Cboe Canada).
38	*OrderQty*	Y	Copied from order.
39	*OrdStatus*	Y	State of order. 0=New 2=Filled 8=Rejected
40	*OrdType*	N	2=Limit
44	*Price*	N	Copied from order.
54	*Side*	Y	Side of the order. 1=Buy 2=Sell 5=Sell Short
55	*Symbol*	Y	Copied from order.
58	*Text*	N	If present, indicates reason for reject or cancel. See *Text*(58) in New Order Single on page 30 for details.
60	*TransactTime*	Y	GMT date-time that transaction occurred.
63	*SettlmntTyp*	N	Copied from order.
64	*FutSettDate*	N	Copied from order.
100	*ExDestination*	N	C=Cross
103	*OrdRejReason*	N	Only used when *ExecType* is Rejected (8). 0=Exchange option 1=Unknown symbol 2=Exchange closed

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TAG	FIELD NAME	REQ'D	DESCRIPTION
			3=Order exceeds limit 5=Unknown order 6=Duplicate order 8=Stale order
150	*ExecType*	Y	Reason for this Execution Report: 0=New (acknowledgement of new order) 2=Fill
151	*LeavesQty*	Y	Quantity of shares still open for further execution.
375	*ContraBroker*	N	The Broker ID (3-digit identifier) of the contra party on this fill. This tag is present if (and only if) *OrdStatus* is 2 (filled). Note: this is a standalone tag and is not used in conjunction with the *NoContraBrokers* (Tag 382) repeating group.
851	*LastLiquidityInd*	N	Present for fills (*ExecType* (150) = 2 (filled). For crosses, buy fill reports will have value 1 and sell fill reports will have value 2. 1=Trade added liquidity 2=Trade removed liquidity
1724	*LEIOrderOrigination*	N	Copied from order (*BuyOrderOrigination/SellOrderOrigination*)
2883	*LEIRoutingArrangementIndicator*	N	Copied from order (*BuyRoutingArrangement/SellRoutingArrangement*).
6750	*UMIRAccountType*	Y	Copied from order (*BuyAccountType/SellAccountType*).
6751	*UMIRUserID*	Y	Copied from order.
6754	*BasketTrade*	N	Copied from order.
6755	*ProgramTrade*	N	Copied from order.
6757	*Jitney*	N	Copied from order (*BuyJitney/SellJitney*).
6761	*Anonymous*	N	Copied from order.
6763	*UMIRRegulationID*	N	Copied from order (*BuyRegulationID/SellRegulationID*).
6773	*CrossType*	N	Copied from order.
6776	*PrincipalTrade*	N	Present on executions where Broker executed a Principal trade. If present, value is Y.
6783	*NonResident*	N	**(Ignored by MATCHNow)** Copied from order.
6791	*ByPass*	N	Copied from order.
6792	*NCIB*	N	Copied from order.
7729	*ShortMarkingExempt*	N	Values will be "0" if either: *Side*= 1 (Buy) and on cross entry *ShortMarkingExempt* was 1 or 3, or if *Side*= 2 (Sell) and on cross entry *ShortMarkingExempt* was 2 or 3. Otherwise, *ShortMarkingExempt* will be absent from the message.
8025	*LEICustomerAccount*	N	Copied from order (*BuyCustomerAccount/ SellCustomerAccount*).
8026	*LEIAlgorithmID*	N	Copied from order (*BuyAlgorithmID/ SellAlgorithmID).*
8027	*CustomerLEI*	N	Copied from order (*BuyCustomerLEI/ SellCustomerLEI).*
8028	*BrokerLEI*	N	Copied from order (*BuyBrokerLEI/ SellBrokerLEI).*
9882	*FeeCode*	N	Specific fee code associated with execution. See the respective market's published fee schedule for possible values. Fee codes may be 1 or 2 alphanumeric characters.
	Standard Message Trailer	Y	

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Implementation Notes

Automatic Cancel on Disconnect or Malfunction

If no messages are received from the Member for two heartbeat intervals, all open orders for that Member will be cancelled automatically. This is done to prevent orders from being stuck in an unknown state in the event of telecommunications failure. Members should choose their heartbeat interval carefully based on the latency and reliability of their telecommunications channel. The minimum supported interval is 5 seconds, and this is also the recommended interval if the latency and reliability of your telecommunications channel support it. Execution Reports for the automatically cancelled orders are available upon reconnection. Members are responsible for rerouting orders to other market centers based on their business needs. This should be rare, but all open orders may also be cancelled in the event of a complete or partial system malfunction.

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Independent Software Vendor (ISV) Configuration

ISVs require special configuration. *ClOrdId* values are required to be unique only within a given *OnBehalfOfCompId*. Execution Report and Cancel Reject messages sent by Cboe Canada will have the *DeliverToCompId* set. **Orders must be cancelled or replaced using the same *OnBehalfOfCompId* as was sent on the Order**.

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FIX Drop

Cboe Canada offers two types of FIX Drop ports (Standard FIX Drop and Order by Order FIX Drop). Neither port type accepts orders. Their purpose is to provide real time information about order flow. They may be configured to send order flow based on various combinations of information relating to specific subscriber firms and/or sessions. With proper authorization (e.g., clearing or sponsored relationships), a single FIX Drop session can be used to obtain information about multiple Members.

Standard FIX Drop

Standard FIX Drop ports only send execution information (i.e., Execution Reports where ***ExecType*** (150) is Partially Filled (1) or Filled (2). Refer to Execution Reports on page 41 for details on standard execution reports.

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Order by Order FIX Drop

Order by Order FIX Drop ports are designed to send more than execution information.

All order message types are supported including, but not limited to Acknowledgements (150=0), Partially Filled (150=1), Filled (150=2), Cancelled (150=4), Replaced (150=5), Rejected (150=8), Order Cancel Rejects (35=9) and optionally (if configured at the port level) Trade Breaks (35=UCC). If the Rejects/Cancels are due to incomplete clearing information, they may be unavailable on Order by Order FIX Drop ports.

Users of Order by Order FIX Drop must always be prepared to receive new/unknown FIX tag and FIX tag values for BOE/FIX ports being monitored. Cboe Canada reserves the right to add new FIX tags and to update values distributed on Order by Order FIX Drop with no notice.

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FIX Drop Port Attributes

Unless specified, both types of FIX Drop ports can be configured with the following features:

Table 25. FIX Drop Port Attributes

ATTRIBUTE	DEFAULT	DESCRIPTION
Send Trade Breaks	Yes	Enables **`Trade Cancel/Correct`** messages (35=UCC). Please note that enabling these messages on Order by Order FIX Drop port will be dependent on **`Trade Cancel/Correct`** messages remaining enabled on corresponding FIX order entry ports.
Unique Wash Execution Ids	No	Appends a .B or .S to *ExecID* (17) on all trades.
Concatenate *CompId* and *SubId*	No	Requires all FIX traffic to contain concatenated (combined) *CompId* and *SubIds*.
Send *OrigCompID*/ *OrigSubID*	No	Send *OrigCompID* (9688) and *OrigSubID* (9689).
Send *Account*	Yes	Send *Account* (1).
Send *OrdType*	Yes	Send *OrdType* (40). **Standard FIX Drop only**. Order by Order FIX Drop will receive FIX Tag 40 based on FIX order entry port attribute "Echo Tag 40 on Ack".
Send Microseconds (MATCHNow only)	Yes	Display microsecond level timestamp granularity for *TransactTime* (60), *OrigTime* (42) and *SendingTime* (52). These tags default to millis granularity (Cboe Canada NEO is always in microsecond granularity).
Send Wash Trades	Yes	Send trades identified by Cboe Canada system as wash trades. **Standard FIX Drop only**.

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FIX Port Attributes

The FIX Port Attributes table lists FIX port attributes that are configurable on the port or firm level. Changes to these attributes can be made by sending a written request to tradedeskca@cboe.com.

Table 26. FIX Port Attributes

ATTRIBUTE	DEFAULT	DESCRIPTION
Allow Short Sales	Yes	Allows or disallows short sales.
Allow Test Symbols Only	No	Allow or disallow orders in non-test symbols.
Maximum Order Size	999,999,999	Maximum number of shares allowed per order.
Maximum Order Dollar Value	Unlimited	Maximum order dollar value per order.
Cancel on ME Disconnect	All	When set to N, this setting allows orders to remain open on a Matching Unit failover. When set to Yes, all open orders associated with a session are immediately cancelled in the event of connectivity loss to a Matching Unit. In any event, if a failover takes longer than five minutes, all orders are cancelled unconditionally. All: cancel all orders (all *TimeInForce* values). Day: cancel all DAY, RHO, and GTT orders (Cboe Canada NEO only) None: disabled (cancel nothing)
Cancel on Disconnect	All	The below options are available for cancelling orders as a result of a session disconnect: All: cancel all orders (all *TimeInForce* values). Day: cancel all DAY, RHO, and GTT orders (Cboe Canada NEO only) None: disabled (cancel nothing)
Send Trade Breaks	Yes	Enables Trade Cancel and Trade Correction Messages (35=UCC).
Resize Minimum Quantities	Yes	When the remaining size on an order is less than the defined *MinQty*, *MinQty* will be automatically set to the remaining size (default), or the order will be cancelled back.
Concatenate CompId and SubId	No	Requires all FIX traffic to contain concatenated (combined) Comp and SubIds.
Cancel on Reject	No	Cancels an order upon a modify reject for that order.
Unique Wash Execution Ids	No	Appends a .B or .S to *ExecID* (17) on all trades.
Echo Tag 40 on Ack	Yes	Return *OrdType* (40) value on FIX Ack. Note that this value will also be returned on Order by Order FIX Drop.
Microsecond Timestamp Granularity (MATCHNow only)	Yes	Display microsecond level timestamp granularity for *TransactTime* (60), *OrigTime* (42) and *SendingTime* (52). These tags default to millis granularity. (Cboe Canada NEO is always in microsecond granularity).
Reject Orders on DROP Port Disconnect	No	Allows Subscriber/Sponsoring Firms to associate a DROP port(s) to an order entry port(s). Once the association has been established, if no DROP ports associated with an order entry port are connected, reject orders on the order entry port until at least one of the DROP port session connections have been established.
Reject Orders on DROP Port Timeout (s)	30	Only applicable for sessions where "Reject Orders on DROP Port Disconnect" has been enabled. When the last associated DROP port for the order entry session has disconnected, the reject/cancel actions will be taken on the order entry session if an associated DROP port has not reestablished its connection in the defined time. Minimum value allowed is 20.

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ATTRIBUTE	DEFAULT	DESCRIPTION
Cancel Open Orders on DROP Port Disconnect	No	Only applicable for sessions where "Reject Orders on DROP Port Disconnect" has been enabled. If all DROP ports associated with an order entry port become disconnected, cancel all open orders on the order entry port.
Session Close Handling	Option #3	Customize message that would normally be sent at the regular market close. Also functional during early close situations. Options supported are as follows: **Send Cancels** (Default) - A Cancel message will be sent for all expired orders. **Suppress Cancels** - No message will be sent, and it is the Member's responsibility to close all expired orders in their system. **Send DoneForDay** - A DoneForDay message will be sent in place of the Cancel message for each order that has expired and is no longer eligible for execution. DoneForDay messages are sent via *OrdStatus* (39) = 3 **and** *ExecType* (150) = 3.
Port Order Rate Threshold	Default = 5,000 msgs/sec Max allowed = 5,000 msgs/sec	The maximum allowed message rate on the session. When the first non-administrative message is received, a one second window begins. During the second no more than 4,999 additional non-administrative messages will be allowed within that window. If the rate is exceeded all new orders in the time window are rejected, modifies are treated as cancels, and cancels are processed.
Symbol Order Rate Threshold	5000 msgs/sec	Functions the same as the Port Order Rate Threshold but is calculated at the symbol level. It is capped by the Port Order Rate Threshold.
Make Conditional Eligible (MATCHNow only)	No	Allows Firms to default LP orders to be eligible to interact with Conditional orders. When set to Yes, *ConditionalFlag* (8002) field on non-OLPP Day orders will be set 2 (eligible to interact with Conditionals).
Default *ExecInst*	N	If this port attribute is set, IOC orders without an *ExecInst* (18) will be assigned: **MATCHNow:** N=Trade at any eligible price (Default). M=Trade at the PNBBO midpoint only. b=Trade at any eligible price within the PNBBO. **Cboe Canada NEO:** N=Trade at any eligible price (Default). M=Trade at the PNBBO midpoint only. p=Trade at the less aggressive of PNBBO midpoint or minimum improvement from the PNBBO.
Default *NoTradeFeat*	None	If this port attribute is set, orders without *NoTradeFeat* (7713) specified will be assigned: **MATCHNow:** None=No self-trade prevention IB=Ignore match (do not trade) EB=Suppress from Tape **Cboe Canada NEO:** None=No self-trade prevention EB=Suppress from Tape U=No Self Trade Protection; *NoTradeKey* (7714) will be ignored. N=Cancel Newest

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ATTRIBUTE	DEFAULT	DESCRIPTION
		O=Cancel Oldest D=Decrement larger / Cancel Smaller
Default *NoTradeKey*	None	If this port attribute is set, orders without *NoTradeKey* (7714) specified will be assigned the value that is set here. Alphanumeric with max length of 6 characters.
Service Bureau Port	No	When set to No only allow Broker ID of the subscriber that owns the port. If set to Yes allow all Broker IDs to be sent on the port.
IOC Sweep Condition (MATCHNow only)	False	When enabled, IOC orders will sweep the BIDS Conditionals book in addition to sweeping the Cboe Canada book.

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Contact Information

Please direct questions or comments regarding this specification to tradedeskca@cboe.com.

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Revision History

DOCUMENT VERSION	DATE	DESCRIPTION
1.0.0	03/21/24	Initial combined version for all Cboe Canada books.
1.0.1	05/02/24	Added value 'N' to tag 18 in New Order Single. Updated "Default ExecInst" port attribute description to include value 'r'. Updated "Order Cancel/Replace Request" to include tag *ExpireTime*. Updated description of *TrueMinQty* in "New Order Single".
1.0.2	08/02/24	Updated Account description in New Order Single to allow for space characters. Corrected ExecInst(18) Odd Lot Liquidity Provider value from "O" to "o" in New Order Single. Added new Section 1.2 - Certification Requirement.
1.0.3	08/16/24	Updated "Cboe Canada" and "MATCHNow" terminology throughout. Changed the expected *TargetCompID* value from "CCAN" to "NEO". Corrected the expected value for "visible" from "v" to "V" for *DisplayIndicator* (9479) in New Order Single.
1.0.4	11/07/24	Removed CrossType from "4.4.1 Execution Report Fields". Added *FutSettDate* and *SettlementType* fields to "4.4.3 Trade Cancel/Correct Message Fields". Fixed cross-reference in "5.2 Cross Orders". Updated descriptions for *BasketTrade*, *ShortMarkingExempt*, *BuyOrderOrigination*, and *SellOrderOrigination* in "5.2.1 Cross Order Entry". Updated *BuyAccountType* and *SellAccountType* to be required fields in "5.2.1 Cross Order Entry". Removed TimeInForce from "5.2.2 Cross Order Execution Report". Updated description of Cancel on ME Disconnect and Cancel on Disconnect port attributes. Added Order Restatements section 1.5.5.
1.0.5	12/20/24	Added Send Microseconds to FIX Drop Port Attributes on page 63(MATCHNow only). Updated Port Order Rate Threshold max to 5,000 in FIX Port Attributes on page 64. Added *OrigTime*, *SymbolSfx*, and *TradeLiquidityIndicator* to Trade Cancel/Correct Message Fields on page 47.
1.0.6	01/15/25	Updated with Cboe Titanium branding.
1.0.7	01/31/25	Removed *ExecType* (150) = 6 and E from Execution Report Fields on page 41. Updated *LastLiquidityInd* (851) description in Cross Order Execution Report on page 57. Clarified *NonResident* (6783) is ignored by MATCHNow.
1.0.8	03/07/25	Added *Anonymous* (6761) to Cross Order Entry on page 54 and Cross Order Execution Report on page 57. Updated *ExpireDate* (432) description in NewOrderSingleMessage on page 30 to indicate GTD orders will be rejected if the date is before the current date. Added reason code 'u' to *Text* (58) in Order Protocol - Cboe Canada to Member on page 41.
1.0.9	04/17/25	Updated *SettlmntTyp* (63) description in Cross Order Entry on page 54, removing the "Required if *ExDestination* (100)=S" statement.
1.0.10	06/30/25	Added *ExecInst* = R and updated *PegDifference* description for Cboe Canada NEO (effective 09/30/25).
1.0.11	10/01/25	Removed 09/30/25 effective dates.

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DOCUMENT VERSION	DATE	DESCRIPTION
1.0.12	10/23/25	Added *CrossType*=N (Net asset value (NAV)) in Cross Order Entry on page 54 (effective 01/13/16).
1.0.13	11/18/25	Updated *BeginSeqNo* description from "auto-generated request ID" to "Sequence number of first message in range to be resent" in Resend Request messages on page 23.

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